

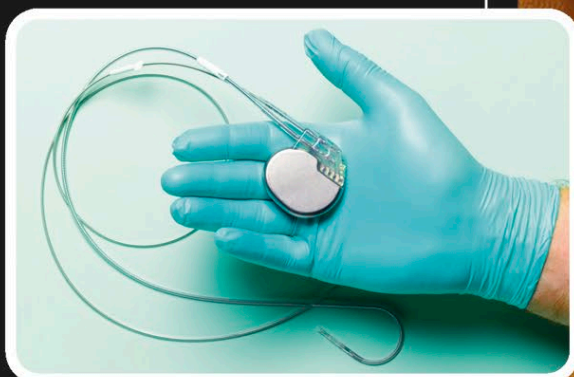
HELPING THE HEART BEAT

PACEMAKERS

The **pacemaker** is a lifesaving invention that helps control the human heartbeat. Millions of people around the world suffer from some form of irregular heartbeat, which can lead to health problems. A pacemaker is a small, battery-powered device that sits inside the body and senses the heartbeat, correcting it when it goes out of rhythm by creating tiny pulses of electricity.

TINY TECH

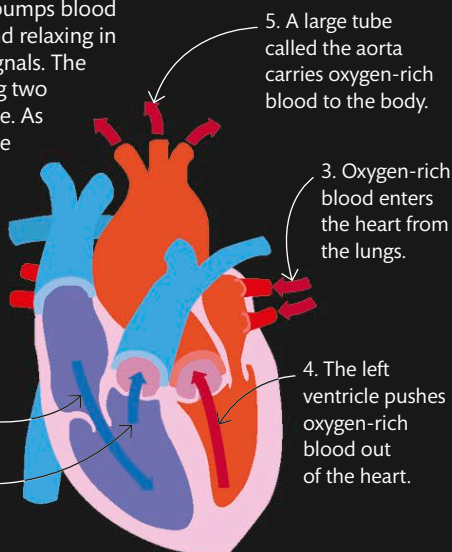
Pacemakers need to be very small to avoid interfering with the body, so most are around the size of a matchbox. Scientists are developing even smaller ones, some as tiny as a pill.



The battery in a pacemaker can last for about **10 years.**

THE HEART

The heart is a powerful organ that pumps blood through the body by contracting and relaxing in a rhythm controlled by electrical signals. The heart has two sides, each containing two chambers—an atrium and a ventricle. As the heart beats, it sends blood to the lungs, where it picks up oxygen. The heart then sucks this blood back, and pumps it out to the rest of the body.



1. Oxygen-poor blood from the body arrives in the right atrium.
2. The right ventricle pushes blood out to the lungs.

5. A large tube called the aorta carries oxygen-rich blood to the body.

3. Oxygen-rich blood enters the heart from the lungs.
4. The left ventricle pushes oxygen-rich blood out of the heart.

➡ OXYGEN-POOR BLOOD
➡ OXYGEN-RICH BLOOD